

Genetic approaches for studying microsporidia

Aaron W. Reinke^{1#}

¹Department of Molecular Genetics, University of Toronto, Toronto, ON, Canada.

Corresponding author

aaron.reinke@utoronto.ca

Abstract

Genetics is the most fundamental technique for understanding gene function. Microsporidia are a large group of parasites that infect most animals. Despite their discovery over 160 years ago, the function of almost all their genes remains undetermined, owing to the difficulty of applying genetic approaches to microsporidia. Within the last decade, however, there has been an increase in studies using such approaches. In this article, I discuss the three major methodologies used so far: RNAi, genome modification, and experimental evolution. RNAi is the most widely used, but its specificity and mechanism in microsporidia remain uncertain. Transformation studies have been reported but have not yet produced a broadly adopted system. Experimental evolution has been applied in diverse host-parasite systems and is emerging as a promising route to forward genetics. For each approach, I describe its history, current evidence, limitations and future steps needed to make genetic analysis of microsporidia more robust.

Keywords: microsporidia; genetics; RNA interference; genome modification; transformation; experimental evolution; intracellular parasites; functional genomics

1. Introduction

Genetics has been a transformative approach to studying biological function [1]. This sentiment is exemplified by Nobel laureate Sydney Brenner who said “Genetics is the master science of biology. In fact it's the only science and all the others are ways of getting to understand what the genes do” [2]. Forward genetic screens have identified genes underlying essential processes and defined genetic interactions and pathways [3–5]. With the advent of genome assemblies, reverse genetic techniques such as knockout collections, RNAi, and CRISPR have enabled systematic interrogation of gene function across diverse phenotypes [6–8]. Genomic modification allows precise protein tagging to determine protein localization, map protein-protein interactions, and exert temporal and spatial control over protein function [9–11]. Together, these classical approaches and newer advances give genetics the potential to greatly accelerate our biological understanding of organisms.

Microsporidia are a large group of intracellular eukaryotic parasites [12,13]. They infect most types of animals, including humans and agriculturally important species [14,15]. They cause disease, reduced lifespan, and death, though they can also have beneficial effects on organisms [16]. Although microsporidia are considered fungi and share some similarities, such as chitin-containing spore walls, they differ from fungi in important ways, including a unique invasion apparatus called the polar tube [17]. They have been difficult to study, and much progress has

relied on biochemical and genomic approaches [18,19]. Although the potential value of applying genetic approaches to microsporidia has been recognized, it has not yet been realized.

Microsporidia have the smallest known eukaryotic genomes, with some encoding fewer than 2,000 proteins, having lost many genes, including those involved in metabolism and gene regulation. Many microsporidian proteins are also shorter than their orthologs in budding yeast [20]. Approximately 800 microsporidian genes show recognizable conservation with other eukaryotes, whereas many predicted genes lack identifiable orthologs outside microsporidia or closely related species [13]. Microsporidia have relatively long branch lengths compared to their fungal relatives, indicative of their highly divergent nature [21]. This led to the assumption that microsporidia have a high mutation rate, but this has not been demonstrated, and the proposed lack of DNA repair machinery instead reflects that divergent DNA repair proteins cannot be detected using sequence-based approaches [22]. Their genomes range from 2.3 to 51.3 Mb, with 6–14 chromosomes [13,23]. Most microsporidia genomes are diploid, but some are tetraploid, a state that has evolved multiple times [24]. Microsporidia are thought to undergo sexual recombination and have conserved the machinery to do so, though this has not yet been observed in a laboratory setting [25,26]. Because many microsporidian genes lack informative homology, sequence-based annotation alone is insufficient for assigning function, making experimental genetic approaches especially valuable [27].

In this review, I discuss the three main genetic approaches that have been applied to studying microsporidia (Figure 1). For each approach, I curated studies applying the technique to microsporidia. I then review and analyze how RNAi has been used to study gene function in microsporidia, describe attempts at genomic modification and transformation, and finally describe experimental evolution and forward genetic approaches. Throughout, I synthesize the studies using these methodologies and offer suggestions on how each could be improved in future work.

2. RNAi

RNA interference (RNAi) is a conserved mechanism of post-transcriptional gene silencing first mechanistically demonstrated in *Caenorhabditis elegans*, with experiments revealing that double-stranded RNA (dsRNA) triggers potent and sequence-specific knockdown of gene function [28,29]. During RNAi, the RNase III enzyme Dicer processes long dsRNA into ~21–25 nucleotide siRNAs, which are loaded onto the Argonaute-containing RNA-induced silencing complex (RISC) that binds to complementary transcripts, silencing the corresponding gene [30]. The first systematic screen for gene function in an animal was performed using RNAi, by feeding *C. elegans* bacteria expressing dsRNAs [31]. RNAi machinery is broadly present in eukaryotes, and knockdown of gene function via RNAi has been performed in a variety of organisms, including intracellular eukaryotic parasites [32,33].

RNAi is the most widely adopted genetic approach used to study microsporidia. This approach is potentially applicable to many microsporidia species as the RNAi machinery components Dicer and Argonaute are present in about half the microsporidia genomes sequenced so far, spanning five of the six major clades [12,13,34,35]. I identified 46 papers in which RNAi was used to attempt knockdown of microsporidia genes across five different species (Figure 2A and Table S1). The first of these studies, published in 2010, used dsRNA feeding to knock down *Vairimorpha ceranae* ADP/ATP transporters in infected honey bees, reducing both target-gene expression and pathogen load [36]. Since then, a variety of proteins have been targeted ranging from eukaryotic-conserved proteins such as metabolic enzymes to microsporidia-specific spore wall and polar tube proteins (Figure 2B) [37–39]. The use of both dsRNA and siRNA targeting microsporidia genes during infection of cells or animals has been reported and a variety of delivery techniques

have been employed including transfection, feeding, injection, and expression in bacteria [40–44].

To understand the magnitude of the effects observed on microsporidia proliferation, I reanalyzed data from papers in which authors measured the impact of RNAi on growth using spore number or expression levels of several genes (18S rRNA, EhSWP1, or beta-tubulin). Values were extracted from published figures using a generative AI model. For studies that also published raw data, the extracted values agreed closely with those calculated from the raw data ($R^2 = 0.99$, $n = 11$ comparisons from four studies; Figure S1). The median effect was a ~50% reduction across these measurements, with outcomes ranging from near-total blockage of growth to increased growth (Figure 2C). Many studies also reported whether the expression of the target gene was reduced. Comparing reported parasite growth inhibition to target gene levels, I found that in most studies the expression of the targeted gene was decreased, and in the case of beta-tubulin there was a significant correlation between growth inhibition and target gene expression (Figure 2D).

Most studies have not examined knockdown specificity in microsporidia, and only one performed RNA-seq after RNAi of the target gene, reporting that several hundred genes were significantly downregulated, but not the target [45]. Several studies have examined the effects of individual off-target genes, and I curated five examples in which genes besides the target were monitored. In two of these studies, there is a significant correlation between knockdown of the target and off-target genes (Figure 2E). In all of these examples, the authors interpret these results as a specific effect of knocking down the target, but a potential alternative explanation is a lack of target specificity [46,47].

Although growth is by far the most commonly monitored phenotype, I observed 8 non-growth phenotypes (Figure 2F). Three of these are host phenotypes, the extent of which are expected to correlate with infection levels [48,49]. The other five are microsporidia-specific phenotypes, but none of the studies that measured these phenotypes include controls showing that unrelated growth-disrupting genes do not also induce the monitored phenotype [50,51]. Thus, although examination of other phenotypes could be informative for determining specificity of the gene being knocked down, the results presented so far do not distinguish between specific effects due to target knockdown, and non-specific reduction of microsporidia growth.

Although the use of RNAi in microsporidia is a potentially exciting approach, much remains to be done to determine its mechanism and to provide confidence in the technique. The apparent assumption underlying all the microsporidia RNAi experiments so far is that, similar to how RNAi functions in other organisms, specific reduction of the target mRNA leads to decreased microsporidia proliferation [30]. In most published examples, the targeting of microsporidia genes leads to reduction of the mRNA of the target gene and decreased microsporidia growth. However, the specificity of this technique has not yet been determined for microsporidia, and the evidence that does exist suggests there is often some amount of non-specific reduction of microsporidia genes.

Two non-mutually exclusive explanations could account for the microsporidia RNAi results reported so far. One, there is knockdown of the target gene, with some degree of knockdown of off-target genes. Two, there is an immune response to the dsRNA or siRNA, reducing both the levels of the target gene as well as overall parasite levels. There is a precedent for a dsRNA-induced response in honey bees, where a dsRNA that is not specific to a virus causes a decrease in virus levels, and dsRNA activates an immune response [52]. Additionally, the antimicrobial peptide apidaecin is upregulated in *V. ceranae*-infected honey bees treated with RNAi targeting

several spore wall proteins, although the authors interpret this as being a specific response caused by decreasing microsporidia proliferation [53].

To distinguish between specific reduction of microsporidia target genes and a potential immune response, a variety of different dsRNA targeting both essential and non-essential genes, as well as non-targeting control sequences, should be tested by RNA-seq, both with and without infection. If there does appear to be a specific knockdown of a microsporidian mRNA, several controls should be used, including multiple sequences targeting the target gene, and sequences targeting likely non-essential microsporidia genes [54,55]. Additionally, researchers should assay genes whose knockdown yields a predictable phenotype other than reduced growth, such as knocking down a polar tube protein and showing the production of spores lacking a polar tube, as well as including RNAi against unrelated microsporidia genes as controls [56]. Additionally, the temporal nature of RNAi could be employed by knocking down genes expected to be involved in spore development, only after meront formation, to avoid the potential of some microsporidia proteins having multiple functions [57].

3. Transformation and genomic modification

Genetic transformation, in which exogenous DNA is introduced into an organism and either transiently expressed or integrated into the genome, has been successfully applied to many eukaryotic intracellular pathogens, providing a road map for microsporidia [18]. Several experimental variables must be considered for microsporidia genome modification, including the DNA construct, how to deliver it, which life stage to deliver it to, how to select for it, and how to maintain the modification [18]. The DNA construct should minimally include a selective marker, regulatory regions to control expression of the marker, and possibly homology regions to enable integration into the genome. This can be facilitated by the easy availability of genome assemblies and RNA-seq data for most species of microsporidia that are commonly studied [58–61]. Although the regulatory sequences controlling gene expression in microsporidia remain poorly characterized, their often short intergenic regions make the complete upstream regions of genes expressed at high levels throughout the life cycle a reasonable starting point for designing expression constructs [62,63]. Many techniques could be used to deliver DNA into microsporidia, including electroporation, transfection, or biolistics, and could target various stages such as spores, intracellular proliferative stages, or potentially the sporoplasm in vitro [18,64–66]. Several options exist for selective markers that have been used successfully in transformation of other intracellular parasites, including drug resistance genes, fluorescent proteins, and luciferases [67,68].

One of the best examples of how genomic modification can revolutionize a field comes from the apicomplexan parasite *Cryptosporidium parvum* [69]. In 2015, using CRISPR–Cas9 together with nanoluciferase expression, a drug-resistance marker and drug selection, transgenic *C. parvum* was generated and gene knockouts were demonstrated [67]. Since this seminal study, these genetic modification techniques have been applied in many fruitful ways. Gene editing has been used to identify essential genes and characterize parasite effectors [70,71]. The life cycle of the parasite was carefully determined through fluorescent protein tagging and microscopy, alongside cell sorting and mRNA sequencing to define stage-specific expression [72]. A follow-up study using *C. parvum* engineered to express a fluorescent reporter applied single-cell RNA sequencing to define the three intracellular stages of this parasite [73]. These transcriptional studies provided insight into the sexual development of *C. parvum*, and by engineering strains with different drug-selectable markers, genetic crosses could be performed [68,72,73]. These crosses were then used to identify genes responsible for virulence [68,74]. Gene editing has also been used to validate drug targets in *C. parvum* [75,76]. Together, *C. parvum* provides both a road map for

making an intracellular eukaryotic parasite genetically tractable and a demonstration of how transformative these approaches can be for researching a pathogen [18,69,77].

Although genetic modification of microsporidia would be a powerful tool to study these parasites, there have been limited studies on this approach so far. I identified 2 papers and one thesis in which transformation of microsporidia was attempted (Figure 3 and Table S2). The first of these studies is a 2006 thesis using *Encephalitozoon cuniculi* infecting cell lines, in which the authors attempted several approaches to transform microsporidia, including electroporation of both infected cells and parasites released from cells, using GFP as a reporter and drug resistance as a selectable marker. However, no stable transformants were detected [78]. The next study, in 2016, examined the species *Nosema bombycis* in both silkworms and cell lines, using GFP as a reporter. A plasmid was transfected into infected silkworm cells, and fluorescence was reported in spores, which was verified by PCR and Western blot. However, the efficiency and stability of this approach were not determined, and a lack of appropriate controls makes the data difficult to interpret. For example, a negative control of spores from non-transfected cells was not included in the PCR validation experiment [65]. Finally, in 2022, *Nosema bombycis* in cell lines was electroporated with plasmids at the intracellular stage, using both red and green fluorescent proteins as reporters. Some fluorescent spores were observed, but again the lack of untransformed microsporidia negative controls and of reported efficiency makes this study challenging to interpret [64]. These attempts to perform genomic transformation of microsporidia are laudable but none of them have been widely adopted, suggesting a potential lack of robustness to the protocols.

There are several recommendations for improving attempts to transform microsporidia. Foremost, using drug selection combined with fluorescence or luminescence would enable both selection and verification of transformation. Brighter reporters such as nanoluciferase and mNeonGreen have been used in *C. parvum*, which could enable the detection of transformants with low expression [67,72]. Several drug resistance markers have been proposed, including dextrazoxane with topoisomerase II resistance sequences, bleomycin with a bleomycin resistance gene, and *Saccharomyces cerevisiae* MetAP1 as a resistance marker for fumagillin [18,79,80]. The ability to select transformed parasites with drugs would make it possible to enrich them for characterization, such as determining whether DNA has integrated into the genome, which could be assessed through whole-genome sequencing. Additionally, given the widespread success of CRISPR for genomic modification in a wide range of organisms, this approach is worth pursuing in microsporidia [81].

4. Experimental evolution

Experimental evolution is an approach in which organisms are evolved under various conditions [82]. This approach has been around since the late 1800s, but has seen an explosion of interest in many fields over the last several decades [83]. This approach allows for determining adaptations to different environments, dynamics and reproducibility of the evolutionary process, and fitness trade-offs to other environments [84]. This approach has been particularly useful to study microbes given the ease of generating large population sizes and their fast generation times [85]. Experimental evolution has been applied to microbes for over 100 years and has been used in experiments of a few generations over the course of several days to thousands of generations over decades [86]. Recently, due to rapid advances in DNA sequencing, the ability to use experimental evolution as a form of forward genetics to identify causative genes has become apparent [85–89].

Although free-living organisms such as yeast and bacteria have been most frequently used in experimental evolution experiments, there are a few impressive examples of this approach applied to obligate eukaryotic parasites [85]. Using drug resistance as a phenotype, *Plasmodium falciparum* was evolved in the presence of over 100 small-molecule inhibitors, identifying drug targets, resistance-conferring variants in those targets, and mechanisms of multidrug resistance [90,91]. Gene variants responsible for drug resistance have also been identified in other eukaryotic parasites such as *Leishmania* and *Toxoplasma gondii* [92,93]. Other selective conditions have also been used, such as cell adherence in *Babesia bovis* which revealed VESA1a as a major contributor to this process or lab adaptation of *T. gondii* which identified a mutation in a P4 flippase and many transcriptional changes [94,95].

Experimental evolution has been applied to microsporidia to investigate a variety of scientific questions. I defined experimental evolution broadly to include serial-passage experiments in which microsporidia populations were maintained under specified host, environmental, or drug-selection conditions and subsequently phenotyped or sequenced. I identified 15 papers from 1997 to 2026 that meet these criteria (Table S3). These papers encompassed 11 microsporidia species and a variety of hosts (Figure 4A). The first of these studied *Glugoides intestinalis* infecting *Daphnia magna* and showed that evolving the microsporidia under conditions of high host mortality, mimicked by host replacement, did not lead to increased virulence [96]. Other experimental evolution studies on microsporidia looked at a variety of phenotypes including virulence, parasite transmission, passaging on different hosts or strains, and drug resistance (Figure 4B) [97–100].

Several studies also looked at how experimental passage in the native host affected genomic characteristics, showing that *Nematocida ausubeli* did not exhibit any loss of heterozygosity when passaged for 14 weeks in *C. elegans* and *V. ceranae* slightly increased genetic diversity when passaged for two generations in honey bees [26,101]. Transcriptomics applied to *Vavraia culicis* strains selected to have different transmission timing showed that these isolates had different effects on mosquito immune gene transcription [102]. In the only study to use experimental evolution to identify causative genetic variants, *Nematocida parisii* was experimentally evolved in *C. elegans* in the presence of either of two microsporidia inhibitors. Strains were isolated that were specifically resistant to the condition under which they were evolved, identifying variants causing resistance in the target genes, and showing that mitotic recombination likely allows heterozygous variants to become homozygous [80].

There are several ways to make experimental evolution more informative to understand microsporidia gene function. Sequencing evolved strains allows causative mutations to be identified, and using a large population size with many independent replicates can help facilitate this [80,88,89]. The use of strong selective conditions allows for the fixation of genetic changes and leads to more interpretable outcomes [86]. When feasible, evolving adapted strains without a mutagen is preferable, as it makes the causative mutations easier to identify [88]. Sequencing immediately after the phenotype becomes predominant can reduce the accumulation of neutral mutations, which would otherwise diminish the signal [89]. Although causative changes can be determined from a population under strong selective conditions, selecting and sequencing individual clones can improve the ability to connect genotype to phenotype [80,91]. Passaging evolved strains under non-selective conditions is necessary to confirm that phenotypic changes are due to genetic rather than epigenetic variation, and although the first published microsporidia experimental evolution paper explicitly describes this step, it is very often omitted in subsequent microsporidia experimental evolution experiments (Table S3) [96].

Forward genetics is a very promising approach to understand microsporidia gene function. There are several conditions in which experimental evolution should lead to interesting results such as drug resistance, nutrient limitation, or host adaptation, either of novel hosts or mutations in hosts that reduce infectivity [103–105]. In addition to experimental evolution, forward genetic approaches could identify mutants with reduced fitness through phenotypic screens, or through the generation of sequenced mutant collections that are tested individually, as has been done in other organisms [106].

5. Conclusion

Progress has been made in using genetic approaches to study microsporidia, but the field remains far from realizing its promise. At present, RNAi provides the largest body of functional perturbation data but requires stronger specificity controls. Transformation holds great potential, but there are unresolved technical barriers. Experimental evolution offers the clearest current route to forward genetic discovery in genetically intractable microsporidia. The advantages and limitations of these three approaches for studying gene function are compared in Table 1. They are complementary: experimental evolution can implicate a gene without any prior candidate, and the implicated variants could then be confirmed by modifying the genome or by knocking down the gene with RNAi. A major strength of microsporidia research is the diverse species which have been studied. Certain techniques may be more plausible in different species: *C. elegans* may be well suited to experimental evolution and forward genetic approaches, since animals displaying microsporidia-associated phenotypes are easy to isolate individually. Conversely, whole-animal models may be less useful for genome modification, where cell-based systems could be advantageous for DNA delivery. Once these technologies are properly developed, they will enable studies of drug targets, sexual recombination, the life cycle, gene function, host interactions, and host specificity. When fully realized, these approaches will provide insight complementary to the biochemical, structural, cell biological, and host-genetic approaches currently in use. Applying comparative approaches across species will illuminate evolutionary differences between them and enable taxonomic distinction and the discovery of shared and unique microsporidian features.

6. Methods

(a) Literature searches

Literature searches were conducted in April 2026. For the RNAi analysis, an initial set of publications identified in PubMed using the search term ‘microsporidia RNAi’ was expanded using AI-assisted literature searches for additional studies of RNAi or gene silencing in microsporidia. Transformation studies were identified using AI-assisted searches focused on direct genetic manipulation of microsporidia, whereas experimental-evolution studies were identified using searches encompassing experimental evolution, serial passage, adaptation and selection of microsporidian populations. Searches included both general microsporidian terminology and individual parasite or host systems where appropriate. All candidate publications were manually inspected to confirm that they met the inclusion criteria described in sections 2–4.

(b) Quantitative data extraction

Quantitative RNAi effects were extracted from published figures. Treatment and corresponding negative-control values were estimated from published axis scales at each reported time point using AI-assisted figure extraction. At least one extracted value from every study was manually checked against the original figure; for studies contributing multiple data points, several values were manually checked across the extracted dataset to verify treatment and control identities, time points and numerical estimates. For bar plots, central bar values were used, whereas

medians were extracted from boxplots. RNAi effects were standardized as the ratio of the RNAi-treated value to the corresponding control value. Measurements presented on transformed scales were handled on the appropriate mathematical scale before calculating ratios: log2- and log10-transformed values were back-transformed, and Ct-derived qPCR measurements were converted using the appropriate base-2 relationship rather than by direct division. Broken axes were interpreted using the scale segment containing the plotted value. Measures of parasite abundance, including spore number, β -tubulin copy number and 18S rRNA abundance, were retained as separate phenotypes. In one case, raw data were used instead of values extracted from the figure because the data were presented on a logarithmic scale (Fig. 4H,I of [42]).

(c) Figure generation

Figures 2–4 and S1 were generated in R version 4.6.1 [107] using ggplot2 [108] and assembled with patchwork [109].

(d) Statistics

Data were analyzed in R version 4.6.1 [107]. Values from RNAi experiments were expressed as log2-transformed ratios of treatment to control. For each paper, gene and phenotype, only the timepoint with the strongest reduction in phenotype was used. Relationships between target mRNA knockdown and either phenotype or off-target mRNA levels were assessed for each group by ordinary least-squares linear regression. For each regression, the coefficient of determination (R^2) is reported, and the slope was tested against zero by a two-sided t-test at $p < 0.05$ without correction for multiple comparisons. All extracted data are included in Table S1.

Ethics

This work did not require ethical approval from a human subject or animal welfare committee.

Data and code accessibility

All data from the paper are provided in supplementary tables S1–S3. All code used for analysis and figure generation is available at <https://github.com/reinkelab-utoronto/microsporidia-genetic-approaches>.

Declaration of AI use

I used the generative AI models Claude Opus 4, Claude Opus 5.5, ChatGPT 5, and Gemini 3 to assist in creating this article. These models were used to assist with literature review, assembly of draft supplementary tables, analysis of published figures, generation of Figure 1, code for creating Figures 2–4 and S1, generation of draft text for figure legends and methods, and copy-editing. All cited literature, code outputs, figures, interpretations, and text were checked and edited by the author. I take full responsibility for all the content in this article.

Authors' contributions

A.W.R.: conceptualization, data curation, writing—original draft, writing—review and editing.

Conflict of interest declaration

I declare I have no competing interests.

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Theme

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Table 1. Genetic approaches for studying microsporidia gene function. Advantages and potential limitations of the three approaches that have been applied to microsporidia. Limitations are those that currently apply to microsporidia specifically, rather than to the technique in general.

Approach	Advantages	Potential limitations
RNAi	Has been reported to work in five species, and the machinery is present in about half of sequenced genomes; in theory any gene can be targeted; several delivery routes are established, including feeding, injection, transfection and expression in bacteria; the timing of delivery can be varied to separate roles at different life-cycle stages; could in principle be scaled to genome-wide screens.	The mechanism has not been demonstrated in microsporidia, specificity is unestablished, and observed effects may instead reflect a host response to dsRNA or siRNA; knockdown is partial and variable, averaging around a 50% reduction in growth but ranging from near-total inhibition to increased growth; produces no stable or heritable change, and therefore no null alleles.
Transformation and genomic modification	The only approach giving precise, heritable modification of a chosen gene; would enable protein tagging and localization, stage-specific expression analysis, reporters, conditional control of gene function, and definitive tests of gene essentiality.	Not yet convincingly established in microsporidia; the intracellular lifestyle complicates DNA delivery, and which life stage to target is unresolved; no selectable marker or promoter has been validated; diploid, or in some species tetraploid, genomes complicate the recovery of homozygous edits; many genes are likely to be essential, so knockouts will not be obtainable for a substantial fraction of the genome.
Experimental evolution	Applicable to any culturable species, including those that are otherwise genetically intractable; unbiased, requiring no prior candidate; isolation of drug-resistant mutants and identification of the causative variants has been demonstrated in <i>Nematocida parisii</i> .	Limited to phenotypes that can be placed under selection, so gene function is only accessible where its change has a selectable consequence; a chosen gene cannot be targeted; requires strong selection over many passages, and causative variants must be distinguished from neutral and hitchhiking mutations; recessive variants in diploid genomes would require recombination to become homozygous; a candidate variant cannot currently be confirmed by re-introduction.

Figure and Legends

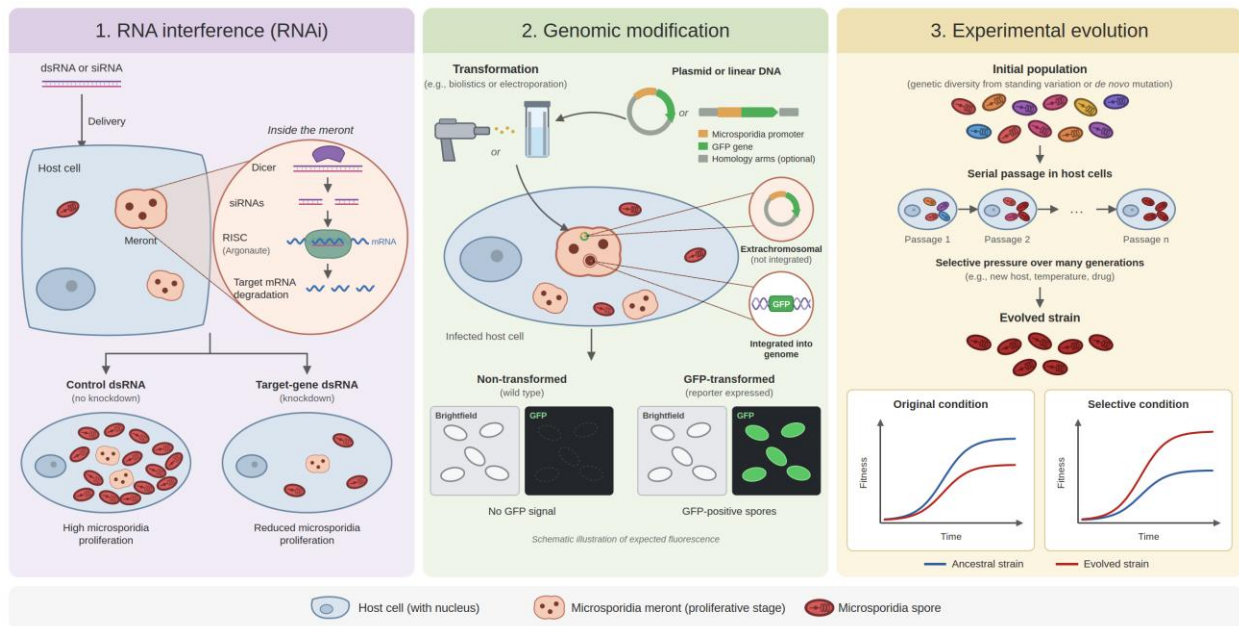


Figure 1. Overview of genetic approaches for studying microsporidia. Schematic overview of three experimental strategies for interrogating microsporidia biology. (1) RNA interference (RNAi). Double-stranded RNA (dsRNA) or small interfering RNA (siRNA) is delivered into infected host cells. Within the microsporidia meront (magnified), Dicer processes dsRNA into siRNAs that are loaded into the RNA-induced silencing complex (RISC), which contains an Argonaute protein, to direct sequence-specific degradation of the target mRNA. Relative to cells treated with a control dsRNA, cells treated with a dsRNA against an essential microsporidia gene show reduced microsporidia proliferation, providing a measure of target-gene function. Note that this schematic is based on the mechanism of RNAi in other organisms and has not been demonstrated for microsporidia. (2) Genomic modification. Exogenous DNA, either a plasmid or linear DNA, is introduced into spores or intracellular stages by transformation (e.g., biolistics or electroporation). The DNA carries a GFP reporter gene under the control of a microsporidia promoter and may be flanked by homology arms. Once inside the parasite, the construct may be maintained as an extrachromosomal plasmid or integrated into the microsporidia genome (magnified insets). Successful transformation can be detected as GFP fluorescence in transformed spores, which is absent from non-transformed (wild-type) spores; the brightfield and GFP panels illustrate the expected result. (3) Experimental evolution is an approach to select for microsporidia carrying genetic variation, arising either from standing variation or *de novo* mutation (spore colours represent different genotypes). The microsporidia population is serially passaged through host cells under a defined selective pressure (e.g., new host, temperature or drug) over many generations, yielding an evolved strain. Fitness of the ancestral and evolved strains over time is then compared under both the original condition and the selective condition to detect adaptive change. The legend at the bottom denotes host cells (with nucleus), microsporidia meronts, and microsporidia spores. This figure was generated as vector graphics using the generative AI model

Claude Opus 5.5 (Anthropic), then checked and edited by the author. All panels are schematic illustrations and do not depict experimental data.

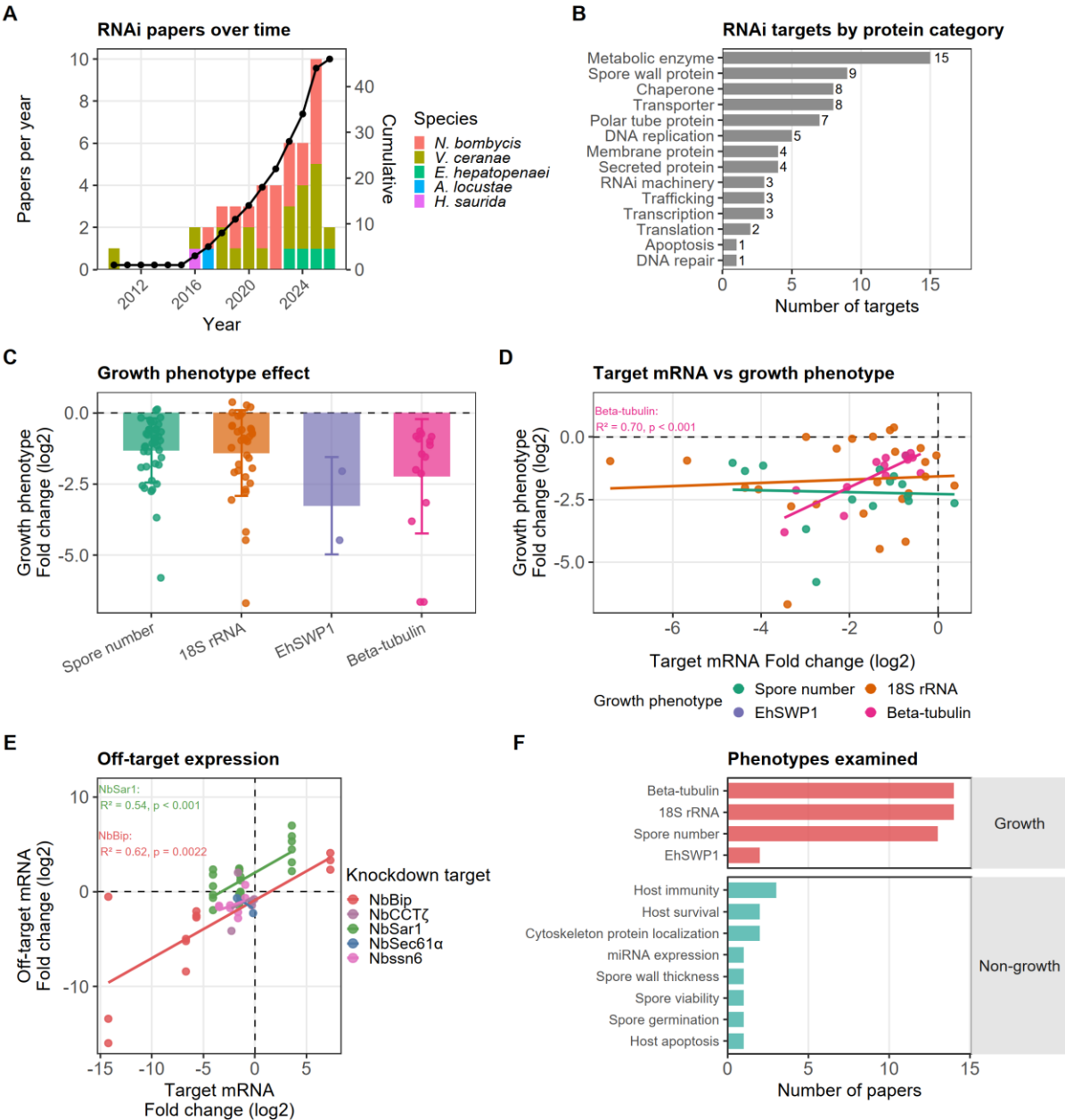


Figure 2. Analysis of microsporidia RNAi studies. (A) Number of microsporidia RNAi papers published per year (stacked bars, colored by parasite species; left axis) and the cumulative total (black line and points; right axis), spanning 2010–2026 ($n = 46$ papers across 5 parasite species). (B) Number of RNAi targets assigned to each likely protein category ($n = 73$ study–target combinations across 14 categories), ordered from most to least frequent. (C) Magnitude of the knockdown phenotype at the most strongly reduced timepoint for each paper–gene–phenotype combination, expressed as log₂ fold-change relative to control. Bars show the mean log₂ fold-

change per phenotype, error bars show ± 1 standard deviation, and points show individual study–gene observations (n = 93 observations from 35 papers: spore number, n = 41; 18S rRNA, n = 35; β -tubulin, n = 15; EhSWP1, n = 2). **(D)** Relationship between target-gene mRNA knockdown and the phenotypic effect at the most strongly reduced timepoint, colored by phenotype (n = 53 paired observations: 18S rRNA, n = 25; spore number, n = 15; β -tubulin, n = 13; no paired mRNA measurements were available for EhSWP1). **(E)** Relationship between knockdown of the intended target mRNA and the mRNA level of co-measured off-target genes, colored by knockdown target (n = 51 target/off-target pairs across 5 knockdown targets: NbSar1, n = 24; NbBip, n = 12; Nbssn6, n = 8; NbCCT ζ , n = 4; NbSec61 α , n = 3). **(D-E)** Regression lines are shown for all phenotypes with sufficient data and (R^2) and slope *p*-value are annotated for phenotypes with a statistically significant relationship of $p < 0.05$. **(F)** Number of papers that examined each phenotype (n = 37 papers), separated into growth-related phenotypes (top, red) and non-growth phenotypes (bottom, teal). All extracted data are available in Table S1.

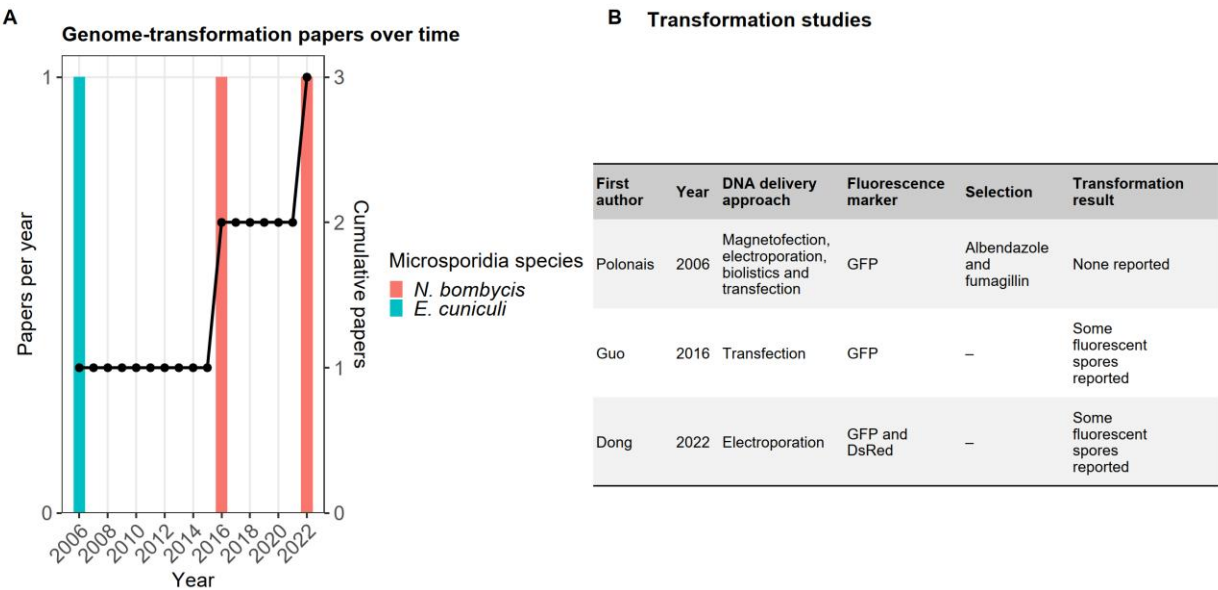


Figure 3. Microsporidia genome-transformation studies. (A) Number of microsporidia genome-transformation studies per year (stacked bars, colored by parasite species; left axis) and the cumulative total (black line and points; right axis), spanning 2006–2022 ($n = 3$ studies across 2 parasite species). (B) Summary of the three reported microsporidia transformation studies, listing first author and year, the approaches used to deliver DNA, the fluorescent reporter, the selection strategy, and the reported outcome.

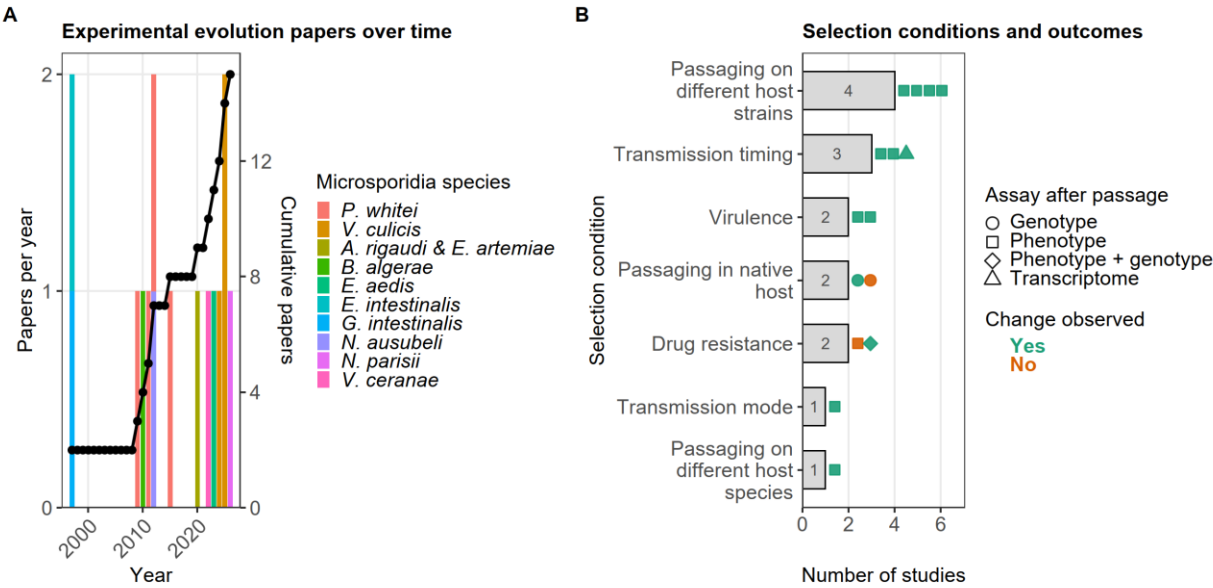


Figure 4. Analysis of microsporidia experimental evolution studies.

(A) Number of microsporidia experimental-evolution and host–parasite coevolution papers published per year (stacked bars, colored by parasite species; left axis) and the cumulative total across all species (black line and points; right axis), spanning 1997–2026 ($n = 15$ papers across 11 parasite species). Bars are stacked by species and ordered in the legend by overall abundance. (B) Number of studies for each selection condition (grey bars). Each point to the right of a bar represents one study, with point shape indicating the assay performed after passage and fill color indicating whether a change in phenotype or genotype after selection was observed (green, yes; orange, no).

Supplementary materials

Table S1. Papers applying RNAi against microsporidia genes.
Table S2. Papers applying transformation and genomic modification to microsporidia.
Table S3. Papers applying experimental evolution to microsporidia.

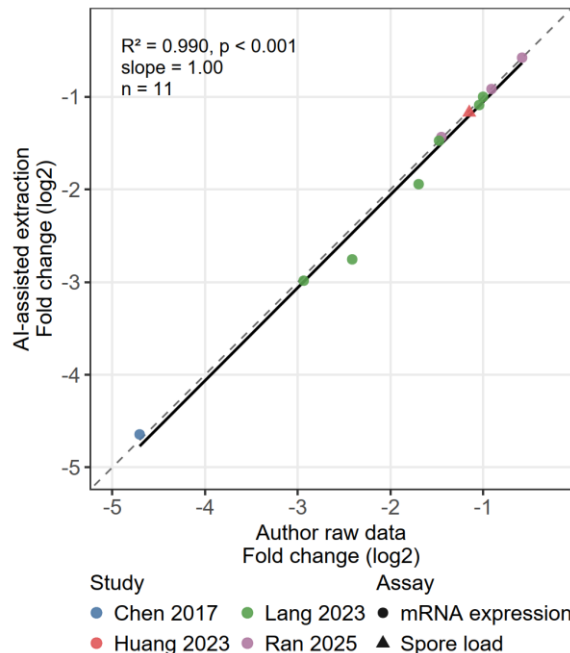


Figure S1. Comparison of AI-assisted figure extraction with author-provided raw data. Fold changes (treatment/control, log2) estimated from published figures with AI-assisted extraction, plotted against the same values calculated from raw data provided by the authors ($n = 11$ comparisons from 4 studies). Points are colored by study, and shape indicates the assay (mRNA expression or spore load). The dashed line shows perfect agreement ($y = x$) and the solid line shows the linear regression fit; R^2 , slope and slope p-value are annotated. Author values are treatment/control ratios of means for mRNA expression and of back-transformed medians for spore load. All values are included in Table S1.

References

1. Nurse P, Hayles J. 2019 Using genetics to understand biology. *Heredity* **123**, 4–13. (doi:10.1038/s41437-019-0209-z)
2. Lawrence PA. 2019 Sydney Brenner: a master of science and of wit. *Development* **146**, dev179879. (doi:10.1242/dev.179879)
3. Forsburg SL. 2001 The art and design of genetic screens: yeast. *Nat Rev Genet* **2**, 659–668. (doi:10.1038/35088500)

- 585 4. Jorgensen EM, Mango SE. 2002 The art and design of genetic screens: *Caenorhabditis*
586 *elegans*. *Nat Rev Genet* **3**, 356–369. (doi:10.1038/nrg794)
- 587 5. St Johnston D. 2002 The art and design of genetic screens: *Drosophila melanogaster*. *Nat*
588 *Rev Genet* **3**, 176–188. (doi:10.1038/nrg751)
- 589 6. Boutros M, Ahringer J. 2008 The art and design of genetic screens: RNA interference. *Nat*
590 *Rev Genet* **9**, 554–566. (doi:10.1038/nrg2364)
- 591 7. Bock C *et al.* 2022 High-content CRISPR screening. *Nat Rev Methods Primers* **2**, 8.
592 (doi:10.1038/s43586-021-00093-4)
- 593 8. Giaever G, Nislow C. 2014 The Yeast Deletion Collection: A Decade of Functional
594 Genomics. *Genetics* **197**, 451–465. (doi:10.1534/genetics.114.161620)
- 595 9. Phanindhar K, Mishra RK. 2023 Auxin-Inducible Degron System: An Efficient Protein
596 Degradation Tool to Study Protein Function. *BioTechniques* **74**, 186–198. (doi:10.2144/btn-
597 2022-0108)
- 598 10. Qin W, Cho KF, Cavanagh PE, Ting AY. 2021 Deciphering molecular interactions by
599 proximity labeling. *Nat Methods* **18**, 133–143. (doi:10.1038/s41592-020-01010-5)
- 600 11. Thorn K, Kellogg D. 2017 Genetically encoded fluorescent tags. *Molecular Biology of the*
601 *Cell* **28**, 848–857. (doi:10.1091/mbc.e16-07-0504)
- 602 12. Bojko J, Reinke AW, Stentiford GD, Williams B, Rogers MSJ, Bass D. 2022 Microsporidia:
603 a new taxonomic, evolutionary, and ecological synthesis. *Trends in Parasitology* **38**, 642–
604 659. (doi:10.1016/j.pt.2022.05.007)
- 605 13. Wadi L, Reinke AW. 2020 Evolution of microsporidia: An extremely successful group of
606 eukaryotic intracellular parasites. *PLoS Pathog* **16**, e1008276.
607 (doi:10.1371/journal.ppat.1008276)
- 608 14. Murareanu BM, Sukhdeo R, Qu R, Jiang J, Reinke AW. 2021 Generation of a Microsporidia
609 Species Attribute Database and Analysis of the Extensive Ecological and Phenotypic
610 Diversity of Microsporidia. *mBio* **12**, 10.1128/mbio.01490-21. (doi:10.1128/mbio.01490-21)
- 611 15. Stentiford GD *et al.* 2016 Microsporidia – Emergent Pathogens in the Global Food Chain.
612 *Trends Parasitol* **32**, 336–348. (doi:10.1016/j.pt.2015.12.004)
- 613 16. Tersigni J, Tamim El Jarkass H, James EB, Reinke AW. 2024 Interactions between
614 microsporidia and other members of the microbiome. *Journal of Eukaryotic Microbiology*
615 **71**, e13025. (doi:10.1111/jeu.13025)
- 616 17. Han B, Takvorian PM, Weiss LM. 2020 Invasion of Host Cells by Microsporidia. *Front.*
617 *Microbiol.* **11**. (doi:10.3389/fmicb.2020.00172)

- 618 18. Reinke AW, Troemel ER. 2015 The Development of Genetic Modification Techniques in
619 Intracellular Parasites and Potential Applications to Microsporidia. *PLoS Pathog* **11**,
620 e1005283. (doi:10.1371/journal.ppat.1005283)
- 621 19. Reinke AW. 2026 Microsporidia: evolution, infection mechanisms and host impact. *Nat Rev*
622 *Microbiol* , 1–16. (doi:10.1038/s41579-026-01364-1)
- 623 20. Jiang J, Qu R, Grigorescu M, Zhao W, Yu QY, Shreenidhi PM, Jarkass HTE, Reinke AW.
624 2025 Functional consequences of reductive protein evolution in a minimal eukaryotic
625 genome. *Cell Reports* **44**. (doi:10.1016/j.celrep.2025.116333)
- 626 21. Rokas A. 2022 Evolution of the human pathogenic lifestyle in fungi. *Nat Microbiol* **7**, 607–
627 619. (doi:10.1038/s41564-022-01112-0)
- 628 22. Mascarenhas dos Santos AC, Julian AT, Pombert J-F. 2022 The Rad9–Rad1–Hus1 DNA
629 Repair Clamp is Found in Microsporidia. *Genome Biology and Evolution* **14**, evac053.
630 (doi:10.1093/gbe/evac053)
- 631 23. Khalaf A *et al.* 2025 Forty new genomes shed light on sexual reproduction and the origin of
632 tetraploidy in Microsporidia. *PLOS Biology* **23**, e3003446.
633 (doi:10.1371/journal.pbio.3003446)
- 634 24. Khalaf A, Lawniczak MKN, Blaxter ML, Jaron KS. 2024 Polyploidy is widespread in
635 Microsporidia. *Microbiology Spectrum* **12**, e03669-23. (doi:10.1128/spectrum.03669-23)
- 636 25. Lee SC, Corradi N, Byrnes EJ, Torres-Martinez S, Dietrich FS, Keeling PJ, Heitman J. 2008
637 Microsporidia Evolved from Ancestral Sexual Fungi. *Current Biology* **18**, 1675–1679.
638 (doi:10.1016/j.cub.2008.09.030)
- 639 26. Cuomo CA *et al.* 2012 Microsporidian genome analysis reveals evolutionary strategies for
640 obligate intracellular growth. *Genome Res* **22**, 2478–2488. (doi:10.1101/gr.142802.112)
- 641 27. Wadi L, El Jarkass HT, Tran TD, Islah N, Luallen RJ, Reinke AW. 2023 Genomic and
642 phenotypic evolution of nematode-infecting microsporidia. *PLOS Pathogens* **19**, e1011510.
643 (doi:10.1371/journal.ppat.1011510)
- 644 28. Sen GL, Blau HM. 2006 A brief history of RNAi: the silence of the genes. *The FASEB*
645 *Journal* **20**, 1293–1299. (doi:10.1096/fj.06-6014rev)
- 646 29. Fire A, Xu S, Montgomery MK, Kostas SA, Driver SE, Mello CC. 1998 Potent and specific
647 genetic interference by double-stranded RNA in *Caenorhabditis elegans*. *Nature* **391**, 806–
648 811. (doi:10.1038/35888)
- 649 30. Wilson RC, Doudna JA. 2013 Molecular Mechanisms of RNA Interference. *Annual Review*
650 *of Biophysics* **42**, 217–239. (doi:10.1146/annurev-biophys-083012-130404)
- 651 31. Kamath RS *et al.* 2003 Systematic functional analysis of the *Caenorhabditis elegans* genome
652 using RNAi. *Nature* **421**, 231–237. (doi:10.1038/nature01278)

- 653 32. Lye L-F, Dobson DE, Beverley SM, Tung M-C. 2025 RNA interference in protozoan
654 parasites and its application. *Journal of Microbiology, Immunology and Infection* **58**, 281–
655 287. (doi:10.1016/j.jmii.2025.01.005)
- 656 33. Cerutti H, Casas-Mollano JA. 2006 On the origin and functions of RNA-mediated silencing:
657 from protists to man. *Curr Genet* **50**, 81–99. (doi:10.1007/s00294-006-0078-x)
- 658 34. Huang Q. 2018 Evolution of *Dicer* and *Argonaute* orthologs in microsporidian parasites.
659 *Infection, Genetics and Evolution* **65**, 329–332. (doi:10.1016/j.meegid.2018.08.011)
- 660 35. de Albuquerque NRM, Ebert D, Haag KL. 2020 Transposable element abundance correlates
661 with mode of transmission in microsporidian parasites. *Mobile DNA* **11**, 19.
662 (doi:10.1186/s13100-020-00218-8)
- 663 36. Paldi N, Glick E, Oliva M, Zilberberg Y, Aubin L, Pettis J, Chen Y, Evans JD. 2010
664 Effective Gene Silencing in a Microsporidian Parasite Associated with Honeybee (*Apis*
665 *mellifera*) Colony Declines. *Applied and Environmental Microbiology*. **76**, 5960–5964.
- 666 37. Rodríguez-García C *et al.* 2018 Nosemosis control in European honey bees, *Apis mellifera*,
667 by silencing the gene encoding *Nosema ceranae* polar tube protein 3. *J Exp Biol* **221**,
668 jeb184606. (doi:10.1242/jeb.184606)
- 669 38. Chen L, Li R, You Y, Zhang K, Zhang L. 2017 A Novel Spore Wall Protein from
670 *Antonospora locustae* (Microsporidia: Nosematidae) Contributes to Sporulation. *J Eukaryot*
671 *Microbiol* **64**, 779–791. (doi:10.1111/jeu.12410)
- 672 39. Dong C, Ma M, Xue R, Ling M, Shi J, Wang Q, Shen Z, Qian H, Zhang Y. 2025 Functional
673 characterization of trehalases 1, 2, and 4 in *Nosema bombycis* (microsporidia). *Microbial*
674 *Pathogenesis* **208**, 107986. (doi:10.1016/j.micpath.2025.107986)
- 675 40. Saleh M, Kumar G, Abdel-Baki A-A, Dkhil MA, El-Matbouli M, Al-Quraishy S. 2016 In
676 Vitro Gene Silencing of the Fish Microsporidian *Heterosporis saurida* by RNA Interference.
677 *Nucleic Acid Therapeutics* **26**, 250–256. (doi:10.1089/nat.2016.0613)
- 678 41. Thepmanee O *et al.* 2023 Molecular and cellular characterization of four putative nucleotide
679 transporters from the shrimp microsporidian *Enterocytozoon hepatopenaei* (EHP). *Sci Rep*
680 **13**, 20008. (doi:10.1038/s41598-023-47114-8)
- 681 42. Lang H *et al.* 2023 Engineered symbiotic bacteria interfering *Nosema* redox system inhibit
682 microsporidia parasitism in honeybees. *Nat Commun* **14**, 2778. (doi:10.1038/s41467-023-
683 38498-2)
- 684 43. Huang Y *et al.* 2018 A secretory hexokinase plays an active role in the proliferation of
685 *Nosema bombycis*. *PeerJ* **6**, e5658. (doi:10.7717/peerj.5658)
- 686 44. Lim S, Yoo H, Han G, Kim W, Kim M. 2025 Target gene screening and validation for
687 RNAi-based suppression of the microsporidian parasite *Vairimorpha* (=Nosema) *ceranae* in
688 honeybees. *Chem. Biol. Technol. Agric.* **12**, 116. (doi:10.1186/s40538-025-00831-w)

- 689 45. Huang Q, Li W, Chen Y, Retschnig-Tanner G, Yanez O, Neumann P, Evans JD. 2019 Dicer
690 regulates *Nosema ceranae* proliferation in honeybees. *Insect Molecular Biology* **28**, 74–85.
691 (doi:10.1111/imb.12534)
- 692 46. Xu S *et al.* 2025 Glucose-regulated protein 78 regulates the subunit-folding of the CCT
693 complex by modulating gene expression and protein interaction in the microsporidian
694 *Nosema bombycis*. *International Journal of Biological Macromolecules* **290**, 138971.
695 (doi:10.1016/j.ijbiomac.2024.138971)
- 696 47. Sun F, Wang R, He P, Wei E, Wang Q, Tang X, Zhang Y, Zhu F, Shen Z. 2022 Sar1
697 Interacts with Sec23/Sec24 and Sec13/Sec31 Complexes: Insight into Its Involvement in the
698 Assembly of Coat Protein Complex II in the Microsporidian *Nosema bombycis*.
699 *Microbiology Spectrum* **10**, e00719-22. (doi:10.1128/spectrum.00719-22)
- 700 48. Kim I-H, Kim D-J, Gwak W-S, Woo S-D. 2020 Increased survival of the honey bee *Apis*
701 *mellifera* infected with the microsporidian *Nosema ceranae* by effective gene silencing.
702 *Archives of Insect Biochemistry and Physiology* **105**, e21734. (doi:10.1002/arch.21734)
- 703 49. Ran M *et al.* 2025 Microsporidian *Nosema bombycis* secretes serine protease inhibitor to
704 suppress host cell apoptosis via Caspase BmICE. *PLOS Pathogens* **21**, e1012373.
705 (doi:10.1371/journal.ppat.1012373)
- 706 50. Liao H *et al.* 2025 Transcription factor Ste12 regulates the spore maturation and proliferation
707 of silkworm microsporidium *Nosema bombycis*. *Pest Management Science* **81**, 3846–3857.
708 (doi:10.1002/ps.8751)
- 709 51. Qi J *et al.* 2021 CCT δ colocalizes with actin and β -tubulin: Insight into its involvement in the
710 cytoskeleton formation of the intracellular parasite *Nosema bombycis*. *Journal of*
711 *Invertebrate Pathology* **184**, 107646. (doi:10.1016/j.jip.2021.107646)
- 712 52. Brutscher LM, Daughenbaugh KF, Flenniken ML. 2017 Virus and dsRNA-triggered
713 transcriptional responses reveal key components of honey bee antiviral defense. *Sci Rep* **7**,
714 6448. (doi:10.1038/s41598-017-06623-z)
- 715 53. He N, Zhang Y, Duan XL, Li JH, Huang W-F, Evans JD, DeGrandi-Hoffman G, Chen YP,
716 Huang SK. 2021 RNA Interference-Mediated Knockdown of Genes Encoding Spore Wall
717 Proteins Confers Protection against *Nosema ceranae* Infection in the European Honey Bee,
718 *Apis mellifera*. *Microorganisms* **9**, 505. (doi:10.3390/microorganisms9030505)
- 719 54. Cullen BR. 2006 Enhancing and confirming the specificity of RNAi experiments. *Nat*
720 *Methods* **3**, 677–681. (doi:10.1038/nmeth913)
- 721 55. Mehlhorn S, Hunnekuhl VS, Geibel S, Nauen R, Bucher G. 2021 Establishing RNAi for
722 basic research and pest control and identification of the most efficient target genes for pest
723 control: a brief guide. *Front Zool* **18**, 60. (doi:10.1186/s12983-021-00444-7)
- 724 56. Echeverri CJ *et al.* 2006 Minimizing the risk of reporting false positives in large-scale RNAi
725 screens. *Nat Methods* **3**, 777–779. (doi:10.1038/nmeth1006-777)

57. Han B *et al.* 2017 The role of microsporidian polar tube protein 4 (PTP4) in host cell infection. *PLOS Pathogens* **13**, e1006341. (doi:10.1371/journal.ppat.1006341)
58. South LR, Hurdeal VG, Fast NM. In press. Genomics and phylogenetic relationships of microsporidia and their relatives. (doi:10.1111/jeu.13051)
59. Wan YC, Troemel ER, Reinke AW. 2022 Conservation of Nematocida microsporidia gene expression and host response in *Caenorhabditis* nematodes. *PLOS ONE* **17**, e0279103. (doi:10.1371/journal.pone.0279103)
60. Jaroenlak P *et al.* 2025 scRNA-seq uncovers the transcriptional dynamics of *Encephalitozoon intestinalis* parasites in human macrophages. *Nat Commun* **16**, 3269. (doi:10.1038/s41467-025-57837-z)
61. He Q, Luo J, Xu J-Z, Wang C, Meng X, Pan G-Q, Li T, Zhou Z-Y. 2020 Morphology and Transcriptome Analysis of *Nosema bombycis* Sporoplasm and Insights into the Initial Infection of Microsporidia. *mSphere* **5**, 10.1128/msphere.00958-19. (doi:10.1128/msphere.00958-19)
62. Peyretailade E *et al.* 2012 Annotation of microsporidian genomes using transcriptional signals. *Nat Commun* **3**, 1137. (doi:10.1038/ncomms2156)
63. Shreenidhi PM, Reinke AW. 2026 25 Years Since the First Microsporidian Genome Assembly, Insights on How Genomics Has Shaped Our Understanding of Parasites with the Smallest Eukaryotic Genomes. *Annual Review of Microbiology* (doi:10.1146/annurev-micro-042424-034850)
64. Dong Z, Gao N, Deng B, Huang X, Hu C, Chen P, Wu Q, Lu C, Pan M. 2022 Stable transformation of fluorescent proteins into *Nosema bombycis* by electroporation. *Parasites Vectors* **15**, 141. (doi:10.1186/s13071-022-05236-4)
65. Guo R, Cao G, Lu Y, Xue R, Kumar D, Hu X, Gong C. 2016 Exogenous gene can be integrated into *Nosema bombycis* genome by mediating with a non-transposon vector. *Parasitol Res* **115**, 3093–3098. (doi:10.1007/s00436-016-5064-8)
66. He Q, Luo J, Xu J-Z, Meng X-Z, Pan G-Q, Li T, Zhou Z-Y. 2020 *In-vitro* cultivation of *Nosema bombycis* sporoplasms: A method for potential genetic engineering of microsporidia. *Journal of Invertebrate Pathology* **174**, 107420. (doi:10.1016/j.jip.2020.107420)
67. Vinayak S, Pawlowic MC, Sateriale A, Brooks CF, Studstill CJ, Bar-Peled Y, Cipriano MJ, Striepen B. 2015 Genetic modification of the diarrhoeal pathogen *Cryptosporidium parvum*. *Nature* **523**, 477–480. (doi:10.1038/nature14651)
68. Shaw S, Cohn IS, Baptista RP, Xia G, Melillo B, Agyabeng-Dadzie F, Kissinger JC, Striepen B. 2024 Genetic crosses within and between species of *Cryptosporidium*. *Proceedings of the National Academy of Sciences* **121**, e2313210120. (doi:10.1073/pnas.2313210120)

69. Kimball A *et al.* 2026 New tools for exploring parasite biology and elucidating host-pathogen interactions in cryptosporidiosis. *Microbiology and Molecular Biology Reviews* **90**, e00148-22. (doi:10.1128/mmbr.00148-22)
70. Guérin A, Roy NH, Kugler EM, Berry L, Burkhardt JK, Shin J-B, Striepen B. 2021 *Cryptosporidium* rhoptry effector protein ROP1 injected during invasion targets the host cytoskeletal modulator LMO7. *Cell Host & Microbe* **29**, 1407-1420.e5. (doi:10.1016/j.chom.2021.07.002)
71. Watson LC *et al.* 2025 Targeted CRISPR screens reveal genes essential for *Cryptosporidium* survival in the host intestine. *Nat Commun* **16**, 7749. (doi:10.1038/s41467-025-63012-1)
72. Tandel J, English ED, Sateriale A, Gullicksrud JA, Beiting DP, Sullivan MC, Pinkston B, Striepen B. 2019 Life cycle progression and sexual development of the apicomplexan parasite *Cryptosporidium parvum*. *Nat Microbiol* **4**, 2226–2236. (doi:10.1038/s41564-019-0539-x)
73. Walzer KA *et al.* 2024 Transcriptional control of the *Cryptosporidium* life cycle. *Nature* **630**, 174–180. (doi:10.1038/s41586-024-07466-1)
74. He W *et al.* 2025 SKSR1 identified as key virulence factor in *Cryptosporidium* by genetic crossing. *Nat Commun* **16**, 4694. (doi:10.1038/s41467-025-60088-7)
75. Ajiboye J *et al.* 2024 Identification of potent and orally efficacious phosphodiesterase inhibitors in *Cryptosporidium parvum*-infected immunocompromised male mice. *Nat Commun* **15**, 8272. (doi:10.1038/s41467-024-52658-y)
76. Vinayak S *et al.* 2020 Bicyclic azetidines kill the diarrheal pathogen *Cryptosporidium* in mice by inhibiting parasite phenylalanyl-tRNA synthetase. *Science Translational Medicine* **12**, eaba8412. (doi:10.1126/scitranslmed.aba8412)
77. Beverley SM. 2015 CRISPR for *Cryptosporidium*. *Nature* **523**, 413–414. (doi:10.1038/nature14636)
78. Polonais V. 2006 Identification de nouvelles protéines du tube polaire et de la paroi sporale chez différentes espèces microsporidiennes. Essais de transfection d'Encephalitozoon cuniculi. phdthesis, Université Blaise Pascal - Clermont-Ferrand II. See <https://theses.hal.science/tel-00695273>.
79. Parrella P, Elikan AB, Kogan HV, Wague F, Marshalleck CA, Snow JW. 2024 Bleomycin reduces *Vairimorpha* (*Nosema*) *ceranae* infection in honey bees with some evident host toxicity. *Microbiology Spectrum* **12**, e03349-23. (doi:10.1128/spectrum.03349-23)
80. Huang Q, Meng X, Zhao W, Pan G, Chen J, Reinke AW. 2026 Forward genetics reveals microsporidian drug targets and recombination-based resistance. *npj Antimicrob Resist* (doi:10.1038/s44259-026-00245-0)

- 798 81. Wang JY, Doudna JA. 2023 CRISPR technology: A decade of genome editing is only the
799 beginning. *Science* **379**, eadd8643. (doi:10.1126/science.add8643)
- 800 82. Buckling A, Craig Maclean R, Brockhurst MA, Colegrave N. 2009 The Beagle in a bottle.
801 *Nature* **457**, 824–829. (doi:10.1038/nature07892)
- 802 83. Geiler-Samerotte K, Lang GI. 2023 Best Practices in Microbial Experimental Evolution. *J*
803 *Mol Evol* **91**, 237–240. (doi:10.1007/s00239-023-10119-y)
- 804 84. Elena SF, Lenski RE. 2003 Evolution experiments with microorganisms: the dynamics and
805 genetic bases of adaptation. *Nat Rev Genet* **4**, 457–469. (doi:10.1038/nrg1088)
- 806 85. Adams J, Rosenzweig F. 2014 Experimental microbial evolution: history and conceptual
807 underpinnings. *Genomics* **104**, 393–398. (doi:10.1016/j.ygeno.2014.10.004)
- 808 86. McDonald MJ. 2019 Microbial Experimental Evolution – a proving ground for evolutionary
809 theory and a tool for discovery. *EMBO Rep* **20**, EMBR201846992.
810 (doi:10.15252/embr.201846992)
- 811 87. Cooper VS. 2018 Experimental Evolution as a High-Throughput Screen for Genetic
812 Adaptations. *mSphere* **3**, 10.1128/msphere.00121-18. (doi:10.1128/msphere.00121-18)
- 813 88. Martínez AA, Lang GI. 2023 Identifying Targets of Selection in Laboratory Evolution
814 Experiments. *J Mol Evol* **91**, 345–355. (doi:10.1007/s00239-023-10096-2)
- 815 89. Schlötterer C, Kofler R, Versace E, Tobler R, Franssen SU. 2015 Combining experimental
816 evolution with next-generation sequencing: a powerful tool to study adaptation from
817 standing genetic variation. *Heredity* **114**, 431–440. (doi:10.1038/hdy.2014.86)
- 818 90. Luth MR *et al.* 2024 Systematic in vitro evolution in Plasmodium falciparum reveals key
819 determinants of drug resistance. *Science* **386**, eadk9893. (doi:10.1126/science.adk9893)
- 820 91. Cowell AN *et al.* 2018 Mapping the malaria parasite druggable genome by using in vitro
821 evolution and chemogenomics. *Science* **359**, 191–199. (doi:10.1126/science.aan4472)
- 822 92. Bhattacharya A *et al.* 2019 Coupling chemical mutagenesis to next generation sequencing
823 for the identification of drug resistance mutations in Leishmania. *Nat Commun* **10**, 5627.
824 (doi:10.1038/s41467-019-13344-6)
- 825 93. Rosenberg A, Luth MR, Winzeler EA, Behnke M, Sibley LD. 2019 Evolution of resistance
826 in vitro reveals mechanisms of artemisinin activity in Toxoplasma gondii. *Proceedings of the*
827 *National Academy of Sciences* **116**, 26881–26891. (doi:10.1073/pnas.1914732116)
- 828 94. Hakimi H, Yamagishi J, Sakaguchi M, Fathi A, Lee JS, Verocai GG, Kawazu S, Asada M.
829 2025 ves1α genes expression is the major determinant of Babesia bovis-infected erythrocytes
830 cytoadhesion to endothelial cells. *PLOS Pathogens* **21**, e1012583.
831 (doi:10.1371/journal.ppat.1012583)

- 832 95. Primo VA, Rezvani Y, Farrell A, Murphy CQ, Lou J, Vajdi A, Marth GT, Zarringhalam K,
833 Gubbels M-J. 2021 The Extracellular Milieu of Toxoplasma's Lytic Cycle Drives Lab
834 Adaptation, Primarily by Transcriptional Reprogramming. *mSystems* **6**, e01196-21.
835 (doi:10.1128/mSystems.01196-21)
- 836 96. Ebert D, Mangin KL. 1997 The Influence of Host Demography on the Evolution of
837 Virulence of a Microsporidian Gut Parasite. *Evolution* **51**, 1828–1837. (doi:10.1111/j.1558-
838 5646.1997.tb05106.x)
- 839 97. Rafaluk C, Gildenhard M, Mitschke A, Telschow A, Schulenburg H, Joop G. 2015 Rapid
840 evolution of virulence leading to host extinction under host-parasite coevolution. *BMC Evol*
841 *Biol* **15**, 112. (doi:10.1186/s12862-015-0407-0)
- 842 98. Katiyar SK, Edlind TD. 1997 In vitro susceptibilities of the AIDS-associated microsporidian
843 Encephalitozoon intestinalis to albendazole, its sulfoxide metabolite, and 12 additional
844 benzimidazole derivatives. *Antimicrobial Agents and Chemotherapy* **41**, 2729–2732.
845 (doi:10.1128/aac.41.12.2729)
- 846 99. Legros M, Koella JC. 2010 Experimental evolution of specialization by a microsporidian
847 parasite. *BMC Evolutionary Biology* **10**, 159. (doi:10.1186/1471-2148-10-159)
- 848 100. Silva LM, Koella JC. 2025 Complex interactions in the life cycle of a simple parasite
849 shape the evolution of virulence. *PLOS Pathogens* **21**, e1013294.
850 (doi:10.1371/journal.ppat.1013294)
- 851 101. Wei X, Zheng J, Evans JD, Huang Q. 2022 Transgenerational genomic analyses reveal
852 allelic oscillation and purifying selection in a gut parasite *Nosema ceranae*. *Front. Microbiol.*
853 **13**. (doi:10.3389/fmicb.2022.927892)
- 854 102. Silva LM. 2024 Selection on time to parasite transmission shapes the host *Anopheles*
855 *gambiae* transcriptional response and suggests immune evasion. , 2024.09.22.613703.
856 (doi:10.1101/2024.09.22.613703)
- 857 103. Van den Bergh B, Swings T, Fauvart M, Michiels J. 2018 Experimental Design,
858 Population Dynamics, and Diversity in Microbial Experimental Evolution. *Microbiology and*
859 *Molecular Biology Reviews* **82**, 10.1128/mmbr.00008-18. (doi:10.1128/mmbr.00008-18)
- 860 104. Tamim El Jarkass H, Mok C, Schertzberg MR, Fraser AG, Troemel ER, Reinke AW.
861 2022 An intestinally secreted host factor promotes microsporidia invasion of *C. elegans*.
862 *eLife* **11**, e72458. (doi:10.7554/eLife.72458)
- 863 105. Willis AR, Zhao W, Sukhdeo R, Wadi L, Jarkass HTE, Claycomb JM, Reinke AW. 2021
864 A parental transcriptional response to microsporidia infection induces inherited immunity in
865 offspring. *Science Advances* **7**, eabf3114. (doi:10.1126/sciadv.abf3114)
- 866 106. Thompson O *et al.* 2013 The million mutation project: a new approach to genetics in
867 *Caenorhabditis elegans*. *Genome Res* **23**, 1749–1762. (doi:10.1101/gr.157651.113)

- 868 107. R Core Team. 2026 *R: A Language and Environment for Statistical Computing*. Vienna,
869 Austria: R Foundation for Statistical Computing. (doi:10.32614/R.manuals)
- 870 108. Wickham H. 2016 *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New
871 York. See <https://ggplot2.tidyverse.org>.
- 872 109. Pedersen TL. 2025 *patchwork: The Composer of Plots*.
873 (doi:10.32614/CRAN.package.patchwork)
- 874
- 875